Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • Grade: 100%

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- **Vector Space Properties**
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min

Vector Space Properties

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A **vector space** V over a field $\mathbb{R}$ is defined by two main operations: vector addition and scalar multiplication. These operations must satisfy certain properties to ensure the structure of a vector space.

1. **Addition**: $+ : V \times V \to V$: This means that the addition operation $+$ takes two elements from V (hence $V \times V$) and maps them to another element in V

2. **Scalar multiplication**: $\cdot : R \times V \to V$: This means that scalar multiplication $\cdot$ takes a scalar from $\mathbb{R}$ (the set of real numbers) and a vector from V (the vector space) and maps them to another vector in V

Properties

1. **Associativity of addition**: $u + (v + w) = (u + v) + w$
2. **Commutativity of addition**: $u + v = v + u$
3. **Identity element of addition**: There exists an element $0 \in V$ such that $v + 0 = v$
4. **Inverse elements of addition**: For each $v \in V$, there exists an element $-v \in V$ such that $v + (-v) = 0$
5. **Compatibility of scalar multiplication with field multiplication**: $(\alpha\beta) \cdot v = \alpha \cdot (\beta \cdot v)$
6. **Identity element of scalar multiplication**: $1 \cdot v = v$
7. **Distributivity of scalar multiplication with respect to field addition**:$(\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v$
8. **Distributivity of scalar multiplication with respect to vector addition**: $\alpha \cdot (u + v) = \alpha \cdot u + \alpha \cdot v$

Examples

- $\mathbb{R}^n$: the space of n-dimensional real vectors
- $\mathbb{R}^{m \times n}$: the space of $m \times n$ real matrices
- $\mathbb{C}$: the space of complex numbers

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